N^2-1 Puzzle Problem

It is an $N \times N$ tile board. contain N^2-1 numbered tiles.

ie, 8 Puzzle Problem mean, there are 9 tiles and 8 tiles are numbered.

-> which are numbered from 1 to 8

-> here, one tile is empty. It is known as empty slot

-> Aim of N^2-1 Puzzle Problem

Transform the arrangement of tiles from Initial arrangement to goal arrangement

8 Puzzle Problem

-> 9 ~~num~~ tiles, 8 numbered tiles

1	2	3
4		6
7	5	8

Initial Arrangement / Initial state

$\longrightarrow$

1	2	3
4	5	6
7	8	

Goal Arrangement / goal state

-> to transform the initial arrangement to final / goal arrangement, we can move the tiles in <u>Left, right, up, Down.</u>

-> Diagonal movement is not possible

→ Each next move is generated based on empty slot positions.

→ The root node become E-node.

→ we can decide which node to become an E-node based on Estimation formula.

→ Estimation formula.

$$\hat{c}(x) = f(x) + \hat{g}(x)$$

where,

$\hat{c}(x)$ = lower bound cost of node 'x'.

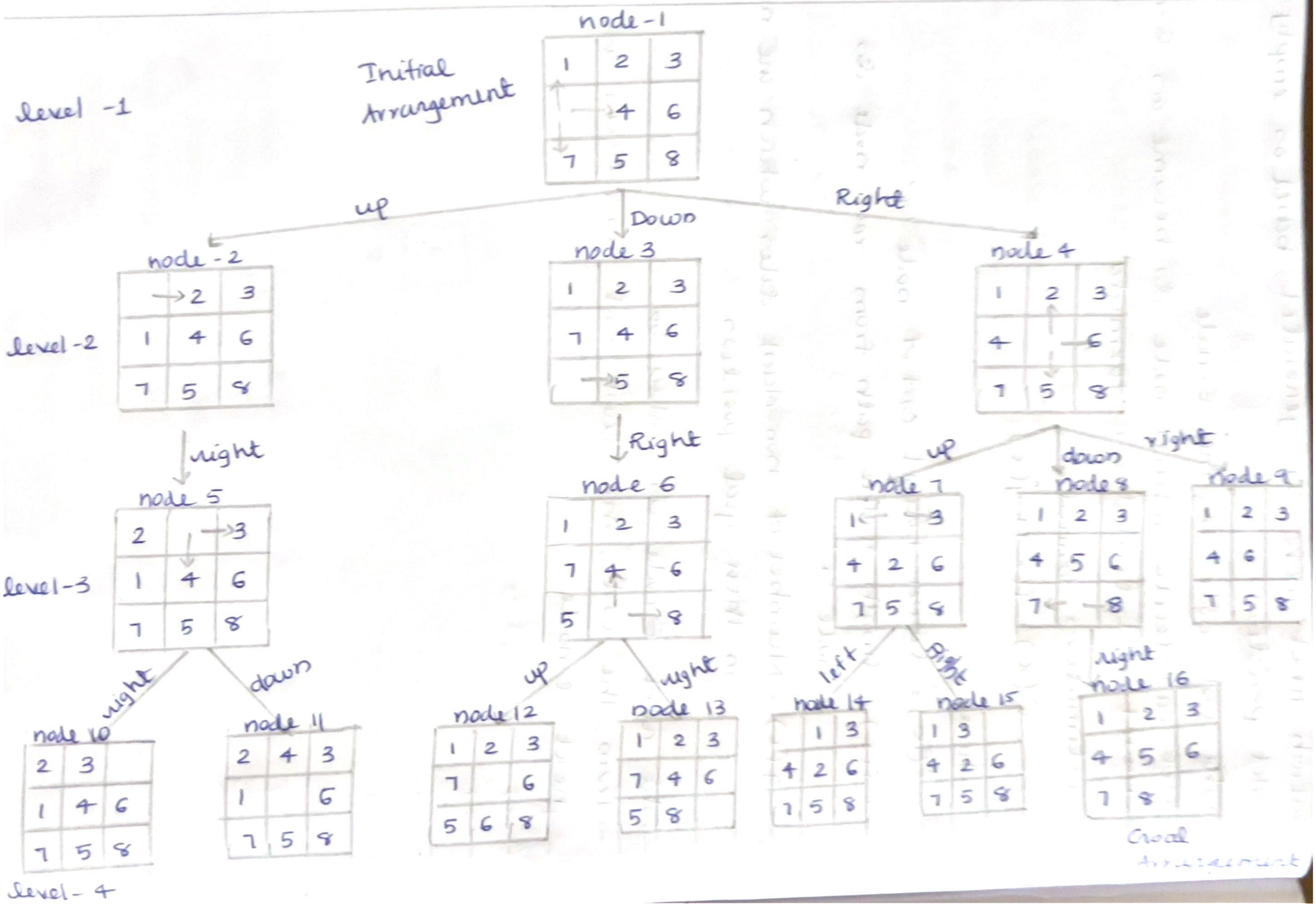
$f(x)$ = length of the path from root node to node 'x'.

$\hat{g}(x)$ = Number of non-blank tiles which are not in their goal position.

→ ~~The next E-node is taken to~~

→ from, the cost, the least cost is taken as the next e-node.

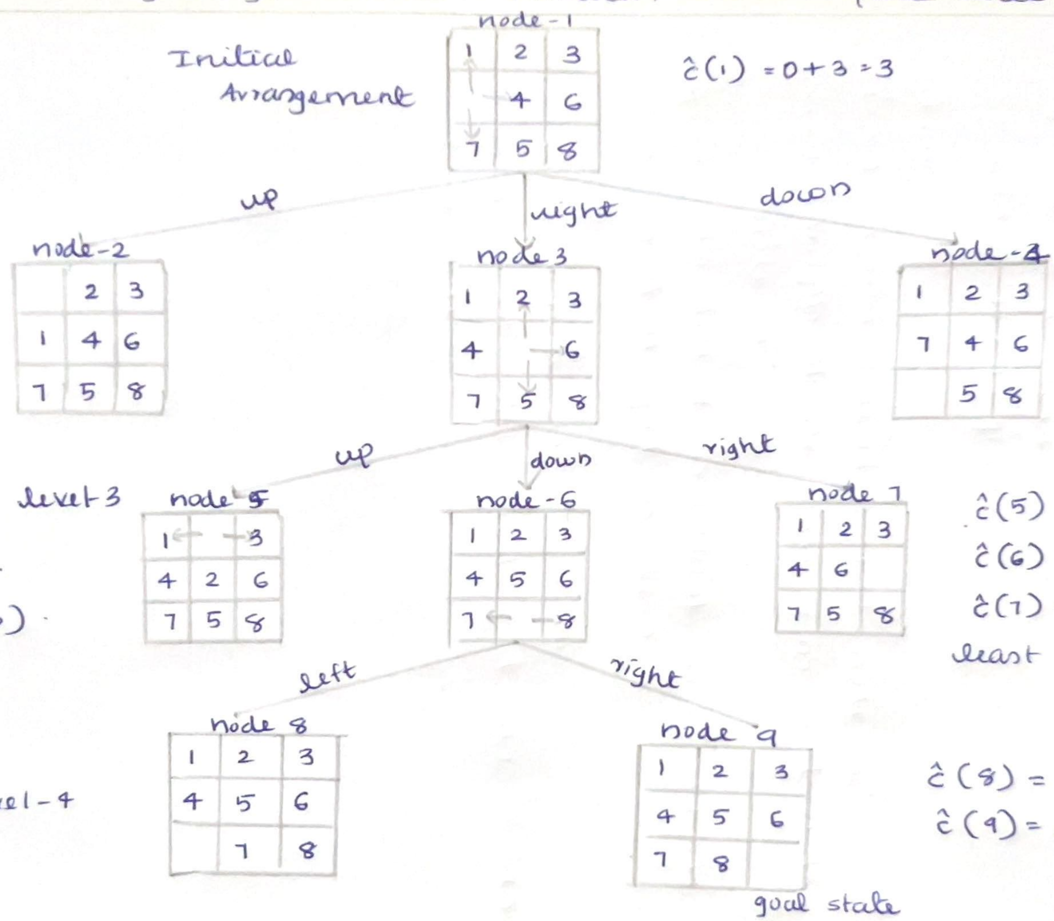
state space tree of 8 puzzle problem



By using Estimation formula, state Space tree

level-1

$\hat{c}(1) = 0+3 = 3$



level-2

$\hat{c}(2) = 1+4 = 5$

$\hat{c}(3) = 1+2 = 3$

$\hat{c}(4) = 1+4 = 5$.

least is $\hat{c}(3)$.

$\hat{c}(5) = 2+3 = 5$

$\hat{c}(6) = 2+1 = 3$

$\hat{c}(7) = 2+3 = 5$.

least is $\hat{c}(6)$.

level-4

$\hat{c}(8) = 3+2 = 5$

$\hat{c}(9) = 3+0 = 3$.

when $\hat{g}(x)$ become 0, then we will get the goal state.